

Homework 1

Due: 2024-3-22 23:59 | 6 Problems, 100 Pts Name: 胡宇阳, ID: 2200013065

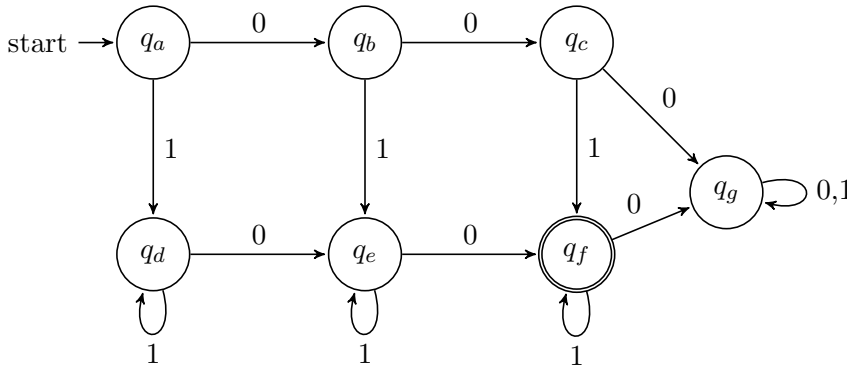
Problem 1.(10 points).

(1) (5 points) Give a DFA recognizing the following language. The alphabet is $\{0, 1\}$.

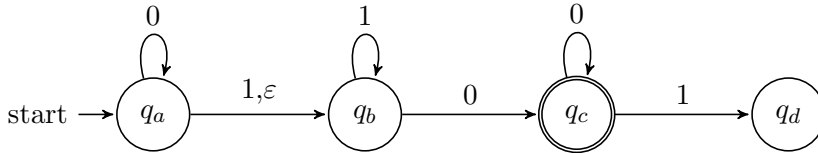
$\{w \mid w \text{ contains exactly two 0s and at least one 1}\}$

(2) (5 points) Give an NFA recognizing the language $0^*1^*0^+$ with four states (including a reject state). The alphabet is $\{0, 1\}$. ◀

Answer. (1)



(2)



Problem 2.(20 points). For any string $w = w_1w_2 \cdots w_n$, the reverse of w , written w^R , is the string w in reverse order, $w_n \cdots w_2w_1$. For any language A , let $A^R = \{w^R \mid w \in A\}$. Show that if A is regular, so is A^R . ◀

Answer. ① 由于 A 为正则语言, 存在 DFA $M = (Q, \Sigma, \delta, q_0, F)$, M 识别 A .

则构造 NFA $\bar{M} = (\bar{Q}, \bar{\Sigma}, \bar{\delta}, \bar{q}_0, \bar{F})$, 满足 $\bar{Q} = Q \cup S_0$, $\bar{\Sigma} = \Sigma \cup \{\varepsilon\}$, $\bar{q}_0 = S_0$, $\bar{F} = q_0$.

$\bar{\delta}$ 是满足 $\forall q_1 \in Q \wedge w \in \Sigma, q_2 = \delta(q_1, w) \Leftrightarrow q_1 \in \bar{\delta}(q_2, w)$ 和 $\forall q \in F, q \in \bar{\delta}(S_0, \varepsilon)$ 的最小集合.

② 下证 $\bar{M}$ 接受 A^R 中的字符串:

$\forall w = w_1w_2 \cdots w_n, w_i \in \Sigma$, 若 w 可被 A 识别, 则 $\exists r_0, r_1, \cdots, r_n \in Q, (r_0 = q_0) \wedge (r_{i+1} = \delta(r_i, w_{i+1}), i = 0, 1, \cdots, n-1) \wedge (r_n \in F)$.

$r_n \in F, \bar{q}_0 = S_0 \Rightarrow r_n \in \bar{\delta}(\bar{q}_0, \varepsilon)$

$r_{i+1} = \delta(r_i, w_{i+1}), i = 0, 1, \cdots, n-1 \Rightarrow r_i \in \bar{\delta}(r_{i+1}, w_{i+1}), i = 0, 1, \cdots, n-1$

$$\bar{F} = q_0, r_0 = q_0 \Rightarrow r_0 \in \bar{F}$$

由 accept 的定义, $\bar{M}$ 识别 A^R .

③ 下证 $\bar{M}$ 拒绝 A^R 外的字符串, 即证 w 被 A^R 接受可以推出 w^R 在 A 中.

$\forall w = w_1 w_2 \cdots w_n, w_i \in \{\Sigma \cup \varepsilon\}$, 若 w 可被 A^R 识别, 则 $\exists r_0, r_1, \dots, r_n \in Q, (r_0 = q_0) \wedge (r_{i+1} \in \delta(r_i, w_{i+1}), i = 0, 1, \dots, n-1) \wedge (r_n \in F)$.

由于 $\bar{M}$ 只有与 S_0 相连的边才识别 ε , 所以只有 $w_1 = \varepsilon, w_2, \dots, w_n \neq \varepsilon$ 才成立. 所以 $w_2 w_3 \cdots w_n$ 可以被 $\bar{M}$ 接受.

$$r_n = \bar{F}, \bar{F} = q_0 \Rightarrow r_n = q_0$$

$$r_{i+1} \in \bar{\delta}(r_i, w_{i+1}), i = 0, 1, \dots, n-1 \Rightarrow r_i = \delta(r_{i+1}, w_{i+1}), i = 1, \dots, n-1$$

$$\forall q \in F, q \in \bar{\delta}(S_0, \varepsilon) \wedge r_1 = \bar{\delta}(S_0, \varepsilon) \Rightarrow r_1 \in F$$

由 accept 的定义, M 接受 $w_n w_{n-1} \cdots w_2$, 证毕. $\triangleleft$

Problem 3.(20 points). Let A be any language. Define DROP-OUT(A) to be the language containing all strings that can be obtained by removing one symbol from a string in A . Thus, DROP-OUT(A) = $\{a \mid \exists y \in \Sigma, \exists x, z \in \Sigma^* : xyz \in A \wedge a = xz\}$. Show that the class of regular languages is closed under the DROP-OUT operation. In other words, if A is a regular language, so is DROP-OUT(A). $\blacktriangleleft$

Answer. ① 由于 A 为正则语言, 存在 DFA $M = (Q, \Sigma, \delta, q_0, F)$, M 识别 A .

则构造 NFA $\bar{M} = (\bar{Q}, \bar{\Sigma}, \bar{\delta}, \bar{q}_0, \bar{F})$, 满足 $\bar{Q} = Q \times \{a, b\}$, $\bar{\Sigma} = \Sigma, \bar{q}_0 = (q_0, a), \forall q \in F, (q, b) \in \bar{F}$.

$\bar{\delta}$ 是满足 $\forall q_1 \in Q \wedge w \in \Sigma, q_2 = \delta(q_1, w) \Leftrightarrow (q_2, a) \in \bar{\delta}((q_1, a), w), (q_2, b) \in \bar{\delta}((q_1, b), w), (q_2, b) \in \bar{\delta}((q_1, a), \varepsilon)$ 的最小集合.

② 下证 $\bar{M}$ 接受 DROP-OUT(A):

设字符串 $A = w_1 w_2 \cdots w_n$ 能被 M 识别, 则 $q_{i+1} = \delta(q_i, w_{i+1})$.

对于 DROP-OUT(A) 中任意一个字符串 $w_1 w_2 \cdots w_{t-1} w_{t+1} \cdots w_n$,

$$(q_i, a) = \bar{\delta}((q_{i-1}, a), w_i), i = 1, 2, \dots, t-1,$$

$$(q_t, b) = \bar{\delta}((q_{t-1}, a), \varepsilon)$$

$$(q_i, b) = \bar{\delta}((q_{i-1}, b), w_i), i = t+1, t+2, \dots, n,$$

所以 $\bar{M}$ 接受 $w_1 w_2 \cdots w_{t-1} w_{t+1} \cdots w_n$

③ 下证被 $\bar{M}$ accept 的字符串在 DROP-OUT(A) 中:

对于被 DROP-OUT(A) accept 的字符串 $w_1 w_2 \cdots w_n$, $\exists$ 对其补充 ε 的方法变为 $x_1 x_2 \cdots x_m, r_0, r_1, \dots, r_m \in \bar{Q}, r_i = \bar{\delta}(r_{i-1}, x_i), i = 1, 2, \dots, m$.

由于从 $(\sim, a)$ 出发, 在 $(\sim, b)$ 结束, 且只有一次从 $(\sim, a)$ 到 $(\sim, b)$ 的机会, 所以 $m = n + 1$ 且 $\exists t, x_1 x_2 \cdots x_m = w_1 w_2 \cdots w_t \varepsilon w_{t+1} \cdots w_n$.

则 $1 \sim t-1$ 时 $r_i = \bar{\delta}(r_{i-1}, w_i), r_i \in (\sim, a)$

$t+1 \sim m$ 时 $r_i = \bar{\delta}(r_{i-1}, w_i)$, $r_i \in (\sim, b)$.

将 NFA 对应回原来的 DFA, $1 \sim t-1$ 时 $q_i = \delta(q_{i-1}, w_i)$,

$t+1 \sim m$ 时 $q_i = \delta(q_{i-1}, w_i)$

设 $q_t = \delta(q_{t-1}, z)$, 则 M 接受 $w_1 w_2 \cdots w_t z w_{t+1} \cdots w_n$.

由于 $w_1 w_2 \cdots w_n$ 在 $\text{DROPOUT}(A)$ 中, $w_1 w_2 \cdots w_t z w_{t+1} \cdots w_n$ 在 A 中. $\triangleleft$

Problem 4.(20 points). For a language A , define an equivalence relation between two strings: $x \equiv y$ means $\forall z \in \Sigma^* : xz \in A \Leftrightarrow yz \in A$. This allows the set Σ^* to be divided into different equivalence classes.

For example, if $A = \{w \mid w \text{ contains exactly two 0s and at least one 1}\}$, then $1 \equiv 1111$, $101 \equiv 110$, but $1 \not\equiv 011$.

(1) (5 points) How many different equivalence classes are there for A in the above example?

(2) (5 points) Prove that for every regular language A , Σ^* can be divided into a finite number of equivalence classes.

(3) (5 points) Prove that if Σ^* can be divided into a finite number of equivalence classes for a language A , then A is a regular language.

(4) (5 points) With the above conclusions, prove $A = \{0^n 1^n \mid n \in \mathbb{N}\}$ is not a regular language. $\blacktriangleleft$

Answer. (1) 7个, 没有1和0, 有1没有0, 没有1有一个0, 有1有一个0, 没有1有两个0, 有1有两个0, 有3个0.

(2) 由于 A 为正则语言, 则存在一个 state 数量最少的 DFA $M = (Q, \Sigma, \delta, q_0, F)$, M 识别 A .

下证任意字符串 $s, \bar{s}$,

由定义, $\exists r_0, r_1, \dots, r_n, \bar{r}_0, \bar{r}_1, \dots, \bar{r}_m \in Q$, $r_0 = \bar{r}_0 = q_0$, $\delta(r_i, w_{i+1}) = r_{i+1}$, $\delta(\bar{r}_i, w_{i+1}) = \bar{r}_{i+1}$

s 和 $\bar{s}$ 在同一个等价类 $\Leftrightarrow r_n = \bar{r}_m$

对于 $q \in Q, z = w_1 w_2 \cdots w_n$, 定义 $\delta^n(q, z) = \delta(\delta(\cdots \delta(q, w_1) \cdots w_{n-1})w_n)$.

① “ $\Leftarrow$ ”

若 $r_n = \bar{r}_m$,

$\therefore \forall z = w_0, w_1, \dots, w_t$, $\delta^t(r_n, z) = \delta^t(\bar{r}_m, z)$

$\therefore \delta^{n+r}(q_0, sz) = \delta^{m+r}(q_0, \bar{s}z)$

$\therefore s, \bar{s}$ 在同一个等价类.

② “ $\Rightarrow$ ”

若 $r_n \neq \bar{r}_m$, 而 s 和 $\bar{s}$ 在同一个等价类, 则 $\forall x \in \Sigma$, $\delta(r_n, x) = \delta(\bar{r}_m, x)$, 可以将 r_n 和 $\bar{r}_m$ 合并为同一个 state, 而不会改变其识别的语言, 与 state 数量最少矛盾.

(3) ① 设 Σ^* 一共 n 个等价类 (用代表元表示) $[x_1], [x_2], \dots, [x_n]$, 则构造一个 NFA $M = (Q, \Sigma, \delta, q_0, F)$ 满足如下条件:

$$Q = \{Q_1, Q_2, \dots, Q_n\}.$$

Σ 不变.

$$q_0 = \{Q_i \mid \varepsilon \in x_i\}$$

$F = \{Q_i \mid \exists x \in Q_i, x \in A\}$, 由于等价类的性质, 取 $z = 0$ 时可得到一个状态对应的等价类只有都不接受和都接受两种状态.

δ 满足 $\forall q_i \in Q, x \in [x_i], w \in \Sigma, [x_iw] \in \delta(q_i, w)$ (通过字符串的拼接进行连接).

② 下证 M 识别 A : 对于 $s = w_1w_2 \dots w_n \in \Sigma^*$, 存在 $r_0, r_1, \dots, r_n \in Q$, $r_i = [w_1w_2 \dots w_i]$, 使得 $r_i = \delta(r_{i-1}, w_i)$.

由 F 的定义, 所以 M 识别 A .

(4) 对于字符串 $s_1 = 0^n1^m, m \leq n, \forall s_2 \in \Sigma^*, s_1s_2 \in A \Leftrightarrow s_2 = 1^{n-m}$.

$\therefore s_1 = 0^n1^m, n - m = C \geq 0$ 在同一个等价类, 且与 $n - m$ 不同的 0^n1^m 字符串不在同一个等价类.

$\therefore 0^11^1, 0^21^1, 0^31^1, \dots$ 均在不同的等价类.

由 (2), A 不是正则语言. ◁

Problem 5.(10 points). Prove that the following languages are not regular.

(1) (5 points) $\{0^m1^n \mid m \text{ and } n \text{ are coprime}\}$.

(2) (5 points) $\{w \mid w \in \{0, 1\}^* \text{ is not a palindrome}\}$. Here a palindrome is a string that reads the same forward and backward. ◀

Answer. (1) 由第四题结论, 将 $A = \{0^m1^n \mid m \text{ and } n \text{ are coprime}\}$ 分为不同的等价类. 对于 0^m1^n , 若 m 为质数, 则 1 的个数必须为 m 的倍数才满足条件,

也就是说对于 $s_1 = 0^m1_1^n, s_2 = 0^m1_2^n, s_1$ 和 s_2 在同一个等价类 $\Leftrightarrow n_1 \equiv n_2 \pmod{m}$.

所以 0^m1^n 有 m 个同余类, 而质数是无限的, 所以有无限个同余类, A 不是正则语言.

(2) 由第四题结论, 将 $A = \{w \mid w \in \{0, 1\}^* \text{ is not a palindrome}\}$ 分为不同的等价类. 对于字符串 $s = 0^n1, z = 10^m, sz$ 为回文串 $\Leftrightarrow m = n$

所以 $01, 001, 0001 \dots$ 均在不同的等价类, 所以有无限个同余类, A 不是正则语言. ◁

Problem 6.(20 points). Let the rotational closure of language A be $RC(A) = \{yx \mid xy \in A\}$.

(1) (10 points) Show that for any language A , we have $RC(A) = RC(RC(A))$.

(2) (10 points) Show that the class of regular languages is closed under rotational closure. In other words, if A is a regular language, so is $RC(A)$. ◀

Answer. (1) 因为 x, y 可以取空集, 所以显然 $RC(A) \subseteq RC(RC(A))$. 下证 $RC(RC(A)) \subseteq RC(A)$.

对于 $str \in RC(RC(A))$, $\exists x, y, z, xyz \in A \wedge (yzx \in RC(RC(A)) \vee zxy \in RC(RC(A)))$

而 $x(yz) \in A \Rightarrow (yz)x \in RC(A)$, $(xy)z \in A \Rightarrow z(xy) \in RC(A)$, $\therefore RC(RC(A)) \subseteq RC(A)$.

$\therefore RC(A) = RC(RC(A))$.

(2) 若 A 为正则语言, 则存在 DFA $M = (Q, \Sigma, \delta, q_0, F)$, M 识别 A .

$\forall q \in Q$, 将 M 拆成两个 DFA $M_{q1} = (Q, \Sigma, \delta, q_0, q)$, $M_{q2} = (Q, \Sigma, \delta, q, F)$, 将 M_{q2} 的接受状态和 M_{q1} 的起始状态用 ε 组合起来, 得到 NFA M_q , 其识别语言 A_q .

下证 $\bigcup_{q \in Q} A_q = RC(A)$

① 对于 $x = w_1 w_2 \cdots w_k, y = w_{k+1} w_{k+2} \cdots w_n, xy \in A$, 则 $\exists r_0, r_1, \dots, r_n \in Q, r_i = \delta(r_{i-1}, w_i)$.

设 $q = r_k$, 由 accept 的定义知 M_{q1} 接受 x , M_{q2} 接受 y , 则 M_q 接受 $yx \in RC(A)$.

所以 $RC(A) \subseteq \bigcup_{q \in Q} A_q$.

② 对于 $s = w_{k+1} w_{k+2} \cdots w_n w_1 w_2 \cdots w_k \in \bigcup_{q \in Q} A_q$, $\exists q, s \in A_q$, 则只需要将 M_{q1} 的接受状态和 M_{q2} 的起始状态用 ε 组合起来, 则可以识别 $w_1 w_2 \cdots w_n \in A$.

所以 $\bigcup_{q \in Q} A_q \subseteq RC(A)$.

综上, $RC(A)$ 也是正则语言.

◁